

Resolution Proof Example.

- (a) Marcus was a man.
- (b) Marcus was a Roman.
- (c) All men are people.
- (d) Caesar was a ruler.
- (e) All Romans were either loyal to Caesar or hated him (or both).
- (f) Everyone is loyal to someone.
- (g) People only try to assassinate rulers they are not loyal to.
- (h) Marcus tried to assassinate Caesar.

Convert to First order Logic.

(a) Marcus was a man.

(a) `man(marcus)`

(b) Marcus was a Roman.

(c) All men are people.

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(h) Marcus tried to assassinate Caesar.

(a) `man(marcus)`

(b) `roman(marcus)`

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- (a) $\text{man}(\text{marcus})$
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- (c) $\forall X. \text{man}(X) \rightarrow \text{person}(X)$

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- (a) `man(marcus)`
- (b) `roman(marcus)`
- (c) $\forall X. \text{man}(X) \rightarrow \text{person}(X)$
- (d) `ruler(caesar)`

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- (h) $\text{tryassasin}(\text{marcus}, \text{caesar})$

Convert to Clausal Form

- (a) $\text{man}(\text{marcus})$
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Convert to Clausal Form

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2. $\text{roman}(\text{marcus})$
3. $(\neg \text{man}(X), \text{person}(X))$
4. $\text{ruler}(\text{caesar})$
5. $(\neg \text{roman}(X), \text{loyal}(X, \text{caesar}), \text{hate}(X, \text{caesar}))$
6. $(\text{loyal}(X, f(X)))$
7. $(\neg \text{person}(X), \neg \text{ruler}(Y), \neg \text{tryassasin}(X, Y), \neg \text{loyal}(X, Y))$
8. $\text{tryassasin}(\text{marcus}, \text{caesar})$

Query

Who hated Caesar?

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In First Order logic.

1. $\exists X. \text{hate}(X, \text{caesar})$

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Who hated Caesar?

In First Order logic.

1. $\exists X. \text{hate}(X, \text{caesar})$

Negate!

1. $\forall X. \neg \text{hate}(X, \text{caesar})$

Query

Who hated Caesar?

In First Order logic.

1. $\exists X. \text{hate}(X, \text{caesar})$

Negate!

1. $\forall X. \neg \text{hate}(X, \text{caesar})$

Clausal Form, with answer literal.

9. $(\neg \text{hate}(X, \text{caesar}), \text{ans}(X))$

Resolution Proof

1. $\text{man}(\text{marcus})$
2. $\text{roman}(\text{marcus})$
3. $(\neg \text{man}(X), \text{person}(X))$
4. $\text{ruler}(\text{caesar})$
5. $(\neg \text{roman}(X), \text{loyal}(X, \text{caesar}), \text{hate}(X, \text{caesar}))$
6. $(\text{loyal}(X, f(X)))$
7. $(\neg \text{person}(X), \neg \text{ruler}(Y), \neg \text{tryassasin}(X, Y), \neg \text{loyal}(X, Y))$
8. $\text{tryassasin}(\text{marcus}, \text{caesar})$
9. $(\neg \text{hate}(X, \text{caesar}), \text{ans}(X))$
10. $R[9, 5c] (\neg \text{roman}(X), \text{loyal}(X, \text{caesar}), \text{ans}(X))$

Note: In general we have to be cautious about variable names. The X in clause 5 is **NOT** the same as the X in clause 9!

Resolution Proof

Note: In general we have to be cautious about variable names. The X in clause 5 is **NOT** the same as the X in clause 9!

1. $(p(X), h(Y))$
2. $(\neg p(X), q(Y))$
3. $R[1,2] (h(Y), q(Y))$

This is incorrect, as now h and q seem to have the same variable. In fact the Y in 1 and the Y in 2 are different. The correct operation is to first rename the variables in one of the clauses so that the two clauses each have distinct variable names.

Resolution Proof

Note: In general we have to be cautious about variable names. The X in clause 5 is **NOT** the same as the X in clause 9!

1. ~~$(p(X), h(Y))$~~ $(p(Z), h(W))$

2. $(\neg p(X), q(Y))$

3. $R[1,2] \{Z=X\} (h(W), q(Y))$

Note that 3 is now more general than $(h(Y), q(Y))$.

In our example if we applied this more correct rule we would still get the same answer...so I took a short cut.

Resolution Proof

1. `man(marcus)`
2. `roman(marcus)`
3. `(¬man(X), person(X))`
4. `ruler(caesar)`
5. `(¬roman(X), loyal(X,caesar), hate(X,caesar))`
6. `(loyal(X,f(X))`
7. `(¬person(X), ¬ruler(Y), ¬tryassasin(X,Y), ¬loyal(X,Y))`
8. `tryassasin(marcus,caesar)`
9. `¬hate(X,caesar)`
10. `R[9,5c] (¬roman(X), loyal(X,caesar), ans(X))`

Resolution Proof

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7. `(¬person(X), ¬ruler(Y), ¬tryassasin(X,Y), ¬loyal(X,Y))`
8. `tryassasin(marcus,caesar)`
9. `¬hate(X,caesar)`
10. `R[9,5c] (¬roman(X), loyal(X,caesar), ans(X))`
11. `R[10a,2] {X=marcus} (loyal(marcus,caesar), ans(marcus))`

Resolution Proof

1. $\text{man}(\text{marcus})$
2. $\text{roman}(\text{marcus})$
3. $(\neg \text{man}(X), \text{person}(X))$
4. $\text{ruler}(\text{caesar})$
5. $(\neg \text{roman}(X), \text{loyal}(X, \text{caesar}), \text{hate}(X, \text{caesar}))$
6. $(\text{loyal}(X, f(X)))$
7. $(\neg \text{person}(X), \neg \text{ruler}(Y), \neg \text{tryassasin}(X, Y), \neg \text{loyal}(X, Y))$
8. $\text{tryassasin}(\text{marcus}, \text{caesar})$
9. $\neg \text{hate}(X, \text{caesar})$
10. $\text{R}[9, 5c] (\neg \text{roman}(X), \text{loyal}(X, \text{caesar}), \text{ans}(X))$
11. $\text{R}[10a, 2] \{X=\text{marcus}\} (\text{loyal}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$
12. $\text{R}[11, 7c] \{X=\text{markus}, Y=\text{caesar}\} (\neg \text{person}(\text{markus}), \neg \text{ruler}(\text{caesar}), \neg \text{tryassasin}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$

Resolution Proof

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2. $\text{roman}(\text{marcus})$
3. $(\neg \text{man}(X), \text{person}(X))$
4. $\text{ruler}(\text{caesar})$
5. $(\neg \text{roman}(X), \text{loyal}(X, \text{caesar}), \text{hate}(X, \text{caesar}))$
6. $(\text{loyal}(X, f(X)))$
7. $(\neg \text{person}(X), \neg \text{ruler}(Y), \neg \text{tryassasin}(X, Y), \neg \text{loyal}(X, Y))$
8. $\text{tryassasin}(\text{marcus}, \text{caesar})$
9. $\neg \text{hate}(X, \text{caesar})$
10. $\text{R}[9, 5c] (\neg \text{roman}(X), \text{loyal}(X, \text{caesar}), \text{ans}(X))$
11. $\text{R}[10a, 2] \{X=\text{marcus}\} (\text{loyal}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$
12. $\text{R}[11, 7c] \{X=\text{markus}, Y=\text{caesar}\} (\neg \text{person}(\text{markus}), \neg \text{ruler}(\text{caesar}), \neg \text{tryassasin}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$
13. $\text{R}[12a, 3b] \{X=\text{markus}\} (\neg \text{man}(\text{markus}), \neg \text{ruler}(\text{caesar}), \neg \text{tryassasin}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$

Resolution Proof

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2. $\text{roman}(\text{marcus})$
3. $(\neg \text{man}(X), \text{person}(X))$
4. $\text{ruler}(\text{caesar})$
5. $(\neg \text{roman}(X), \text{loyal}(X, \text{caesar}), \text{hate}(X, \text{caesar}))$
6. $(\text{loyal}(X, f(X)))$
7. $(\neg \text{person}(X), \neg \text{ruler}(Y), \neg \text{tryassasin}(X, Y), \neg \text{loyal}(X, Y))$
8. $\text{tryassasin}(\text{marcus}, \text{caesar})$
9. $\neg \text{hate}(X, \text{caesar})$
10. $\text{R}[9,5c] (\neg \text{roman}(X), \text{loyal}(X, \text{caesar}), \text{ans}(X))$
11. $\text{R}[10a,2] \{X=\text{marcus}\} (\text{loyal}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$
12. $\text{R}[11,7c] \{X=\text{markus}, Y=\text{caesar}\} (\neg \text{person}(\text{markus}), \neg \text{ruler}(\text{caesar}), \neg \text{tryassasin}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$
13. $\text{R}[12a,3b] \{X=\text{markus}\} (\neg \text{man}(\text{markus}), \neg \text{ruler}(\text{caesar}), \neg \text{tryassasin}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$
14. $\text{R}[13a,1] (\neg \text{ruler}(\text{caesar}), \neg \text{tryassasin}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$

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14. $\text{R}[13a,1] (\neg \text{ruler}(\text{caesar}), \neg \text{tryassasin}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$
15. $\text{R}[13a,2] (\neg \text{tryassasin}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$

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11. $R[10a,2] \{X=\text{marcus}\} (\text{loyal}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$
12. $R[11,7c] \{X=\text{markus}, Y=\text{caesar}\} (\neg \text{person}(\text{markus}), \neg \text{ruler}(\text{caesar}), \neg \text{tryassasin}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$
13. $R[12a,3b] \{X=\text{markus}\} (\neg \text{man}(\text{markus}), \neg \text{ruler}(\text{caesar}), \neg \text{tryassasin}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$
14. $R[13a,1] (\neg \text{ruler}(\text{caesar}), \neg \text{tryassasin}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$
15. $R[13a,2] (\neg \text{tryassasin}(\text{markus}, \text{caesar}), \text{ans}(\text{markus}))$
16. $R[15a,8] \text{ans}(\text{markus})$